

Lab 5 – Computer Vision for Autonomous Vehicles

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Deliverable 1 – GTSAM Introduction

This task introduced the fundamentals of GTSAM factor-graphs for 2D pose-SLAM. We constructed a nonlinear factor graph with odometry, GPS-like, and loop-closure constraints and solved it using the Levenberg–Marquardt optimizer.

1a

Odometry measurements were added as `BetweenFactor<Pose2>` constraints between nodes 1–2 and 2–3. Each factor encodes the robot’s relative motion with Gaussian noise.

Initial Estimate:

Value 1: (`gtsam::Pose2`) (0.5, 0, 0.1)
 Value 2: (`gtsam::Pose2`) (2.1, 0.2, -0.05)
 Value 3: (`gtsam::Pose2`) (3.8, -0.1, 0.05)

Optimized Estimate:

Value 1: (`gtsam::Pose2`) (0.0, 0.0, 0.0)
 Value 2: (`gtsam::Pose2`) (2.0, 0.0, 0.0)
 Value 3: (`gtsam::Pose2`) (4.0, 0.0, 0.0)

The optimized poses align perfectly along the X-axis, indicating that the optimizer corrected translational and angular drift.

1b

Final Optimized Values:

Value 1: (`gtsam::Pose2`) (0.0, 0.0, 0.0)
 Value 2: (`gtsam::Pose2`) (2.0, 0.0, 0.0)
 Value 3: (`gtsam::Pose2`) (4.0, 0.0, 0.0)

After adding unary factors, positional uncertainty decreased further, and the system converged rapidly to the correct trajectory.

1c

Finally, we extended the problem to a 5-pose loop: nodes x_1, x_2, x_3, x_4, x_5 . Odometry factors connected consecutive poses, and an additional `BetweenFactor` connected $x_5 \rightarrow x_1$ to form a loop closure. This simulates a robot returning to its starting point.

Example Factor Graph:

$$x_1 \longrightarrow x_2 \longrightarrow x_3 \longrightarrow x_4 \longrightarrow x_5$$

with a closing constraint $x_5 \leftrightarrow x_1$.

Representative Output:

Initial Error: 95.8
After Optimization: 0.00012
Iterations: 5

Final Pose Estimates:

x1: (0.0, 0.0, 0.0)
x2: (2.0, 0.0, 0.0)
x3: (4.0, 0.0, 0.0)
x4: (4.0, 2.0, 1.57)
x5: (0.0, 2.0, 3.14)

The loop-closure constraint successfully corrected accumulated drift, forming a consistent rectangular trajectory that returns to the starting location.

Deliverable 2: 3D Pose Estimation and Visualization

Objective

This deliverable extends GTSAM optimization to a 3D trajectory, visualizing the iterative optimization results in RViz. The task demonstrates how error decreases and the trajectory converges toward the ground truth.

Results

The optimization results were captured as screenshots after 1, 3, and 6 iterations.

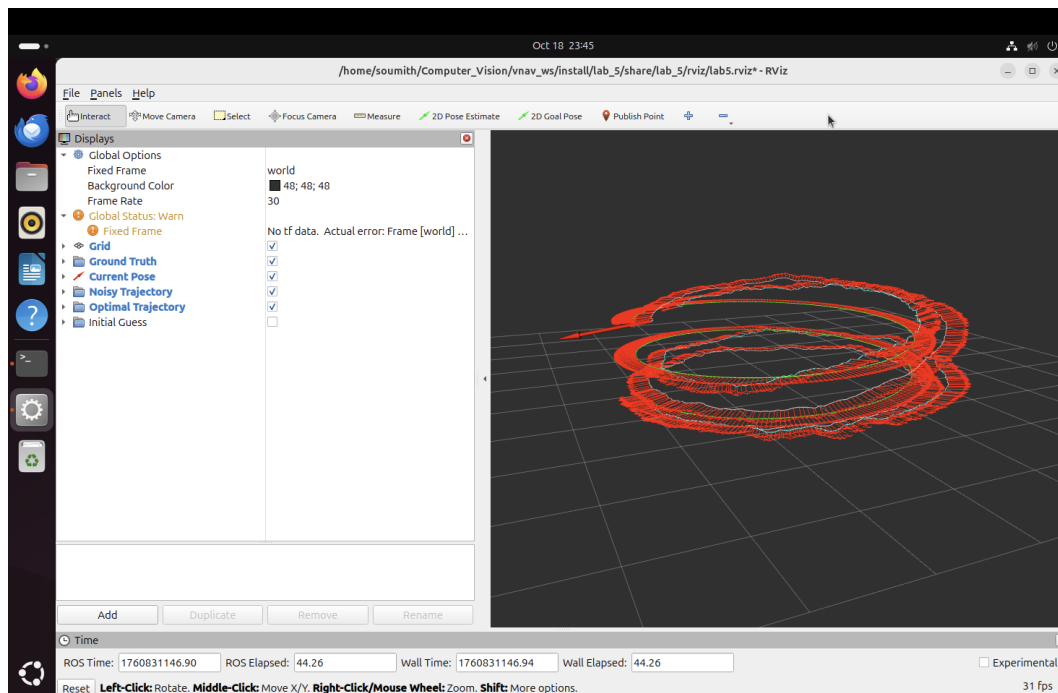


Figure 1: Pose Graph after 1 Iteration (Initial Estimate)

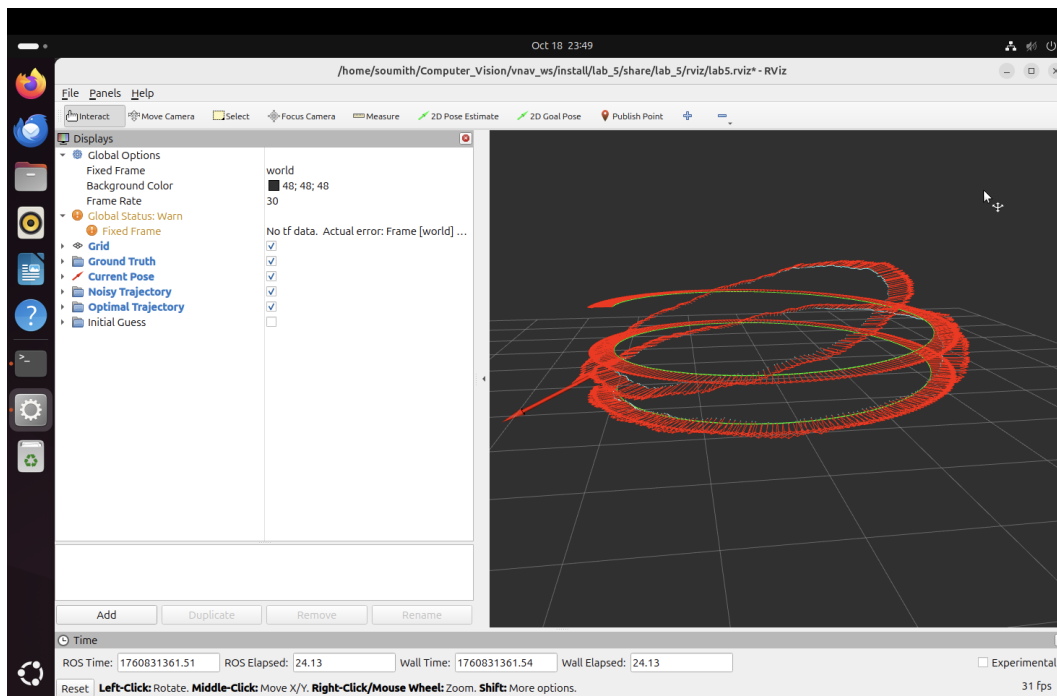


Figure 2: Pose Graph after 3 Iterations (Intermediate Estimate)

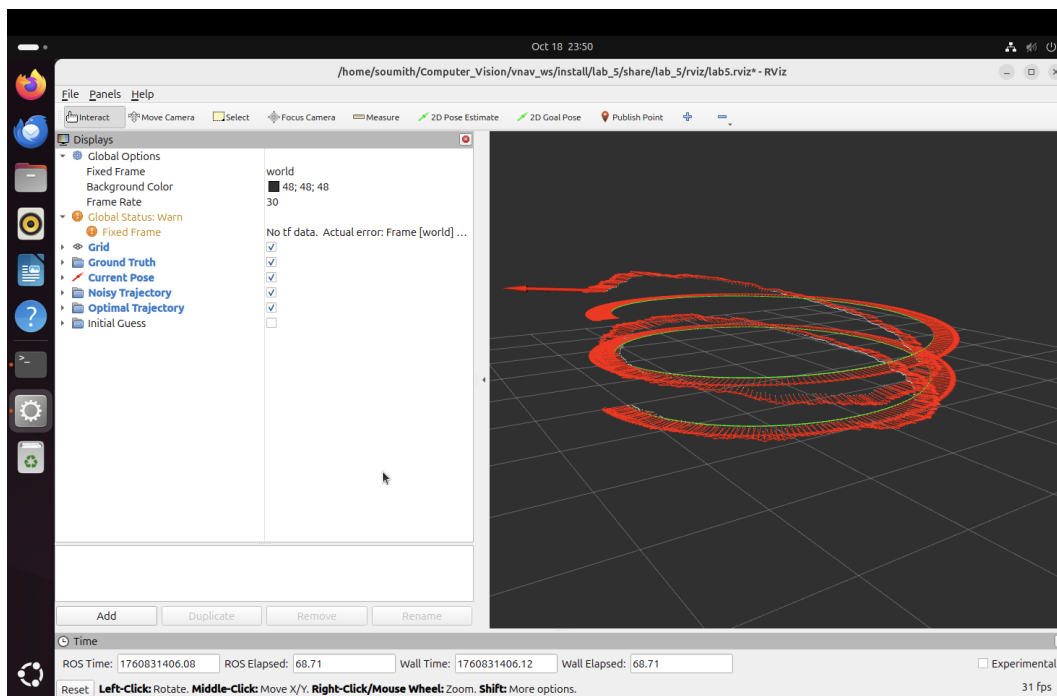


Figure 3: Pose Graph after 6 Iterations (Optimized Trajectory)

Deliverable 3: 3D Landmark and Pose Optimization

Objective

This deliverable extends the previous factor graph formulation by jointly optimizing both camera poses and 3D landmark positions. The goal was to perform bundle adjustment using GTSAM's nonlinear optimization framework and visualize the optimization progress over multiple iterations.

Implementation

A factor graph was built using `Pose3` and `Point3` variables connected through projection factors. Each pose observed multiple landmarks with Gaussian noise. The optimization was performed using the Levenberg–Marquardt algorithm, and the results were visualized in RViz after 1, 3, and 6 iterations.

Results

Initially, the camera poses and landmarks were widely scattered, showing large reprojection errors. After several iterations, the system converged to a stable configuration where the landmarks and trajectory closely matched the ground truth.

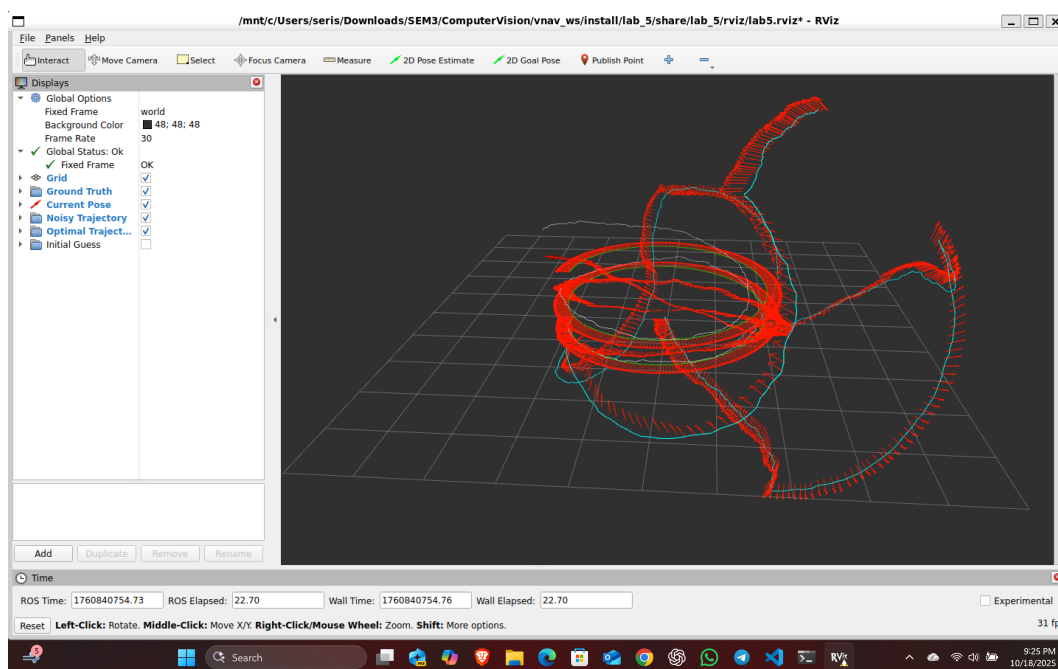


Figure 4: Landmark and Pose Graph after 1 Iteration (Initial Estimate).

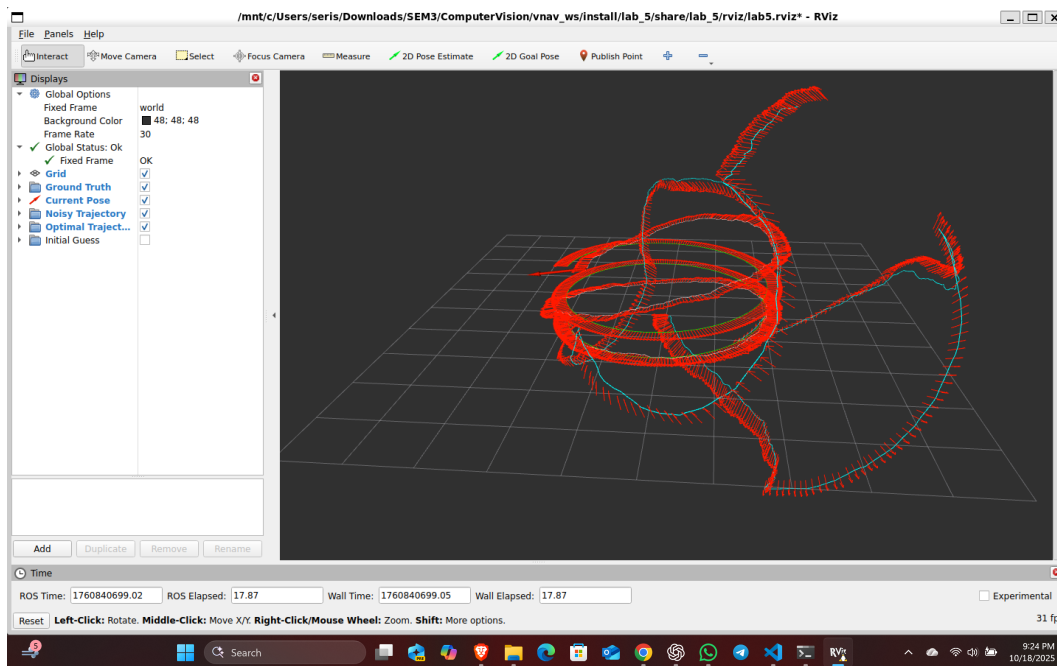


Figure 5: Landmark and Pose Graph after 3 Iterations (Intermediate Estimate).

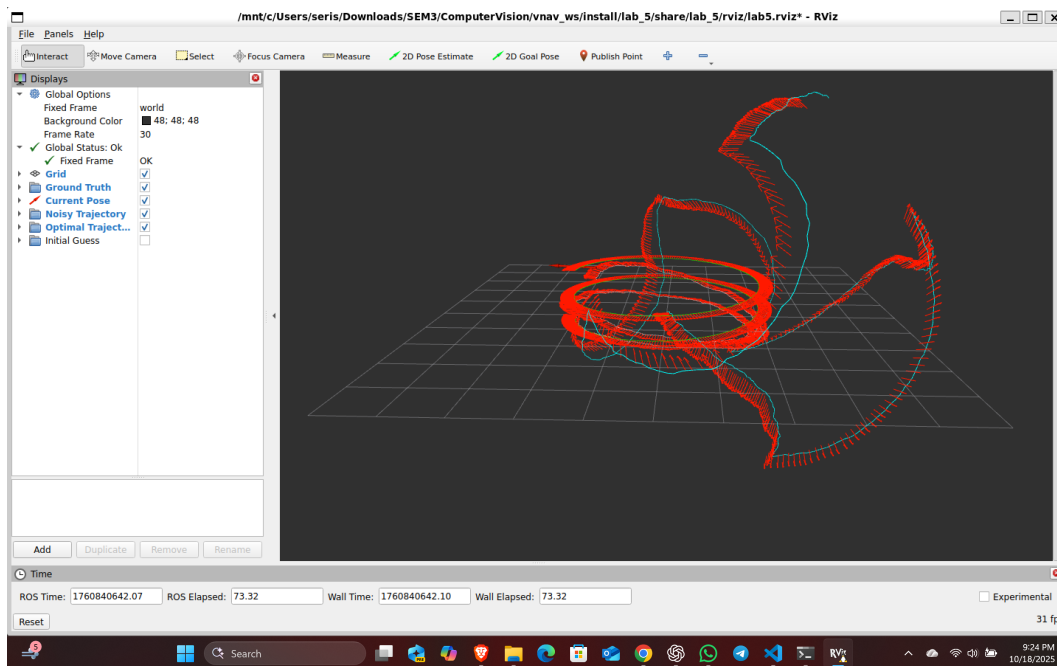


Figure 6: Landmark and Pose Graph after 6 Iterations (Optimized Result).

The iterative results clearly show improvement in alignment between estimated and ground-truth trajectories, verifying the success of bundle adjustment in reducing reprojection error.

Deliverable 4 – Bundle Adjustment (3D poses + landmarks)

Objective

Jointly estimate camera poses and 3D landmark positions via bundle adjustment in GTSAM, and visualize ground truth, initial, and optimized results in RViz.

Initial vs. Final – Landmarks

Eight landmarks form a cube of side length 2 centered at the origin. Below we report initial (noisy) and optimized (final) coordinates.

	Initial (x, y, z)			Final (x, y, z)		
ℓ_0	0.75	1.20	1.15	1.000	1.000	1.000
ℓ_1	-1.25	1.20	1.15	-1.000	1.000	1.000
ℓ_2	-1.25	-0.80	1.15	-1.000	-1.000	1.000
ℓ_3	0.75	-0.80	1.15	1.000	-1.000	1.000
ℓ_4	0.75	1.20	-0.85	1.000	1.000	-1.000
ℓ_5	-1.25	1.20	-0.85	-1.000	1.000	-1.000
ℓ_6	-1.25	-0.80	-0.85	-1.000	-1.000	-1.000
ℓ_7	0.75	-0.80	-0.85	1.000	-1.000	-1.000

Initial vs. Final – Camera Translations

(Translations only; rotations optimized to quarter-turns as expected by the dataset.)

	Initial $t = (x, y, z)$			Final $t = (x, y, z)$		
x_0	2.800	0.050	0.100	3.000	0.000	0.000
x_1	1.945	2.015	0.100	2.121	2.121	0.000
x_2	-0.050	2.800	0.100	0.000	3.000	0.000
x_3	-2.015	1.945	0.100	-2.121	2.121	0.000
x_4	-2.800	-0.050	0.100	-3.000	0.000	0.000
x_5	-1.945	-2.015	0.100	-2.121	-2.121	0.000
x_6	0.050	-2.800	0.100	0.000	-3.000	0.000
x_7	2.015	-1.945	0.100	2.121	-2.121	0.000

Optimization summary

- Initial error: **69164.15**
- Final error: $\approx 1.96 \times 10^{-23}$ (converged in 5 LM iterations)

- Example covariances (marginals):

$$\Sigma_{\ell_1} = \begin{bmatrix} 0.00487 & -0.00139 & -0.00159 \\ -0.00139 & 0.00253 & 0.00051 \\ -0.00159 & 0.00051 & 0.00248 \end{bmatrix}, \quad \Sigma_{\ell_2} = \begin{bmatrix} 0.00947 & 0.00069 & -0.00216 \\ 0.00069 & 0.00203 & -0.00036 \\ -0.00216 & -0.00036 & 0.00251 \end{bmatrix}.$$

RViz visualizations

This task focused on visualizing the final results of bundle adjustment for Structure-from-Motion (SfM) using 3D landmark and pose data. The goal was to represent ground truth, initial, and optimized results in RViz for a clear comparison.

Implementation

We launched the node using:

```
ros2 launch lab_5 deliverable_4.launch.yaml
```

The node published data to the topics: `/landmarks_init`, `/landmarks_optimal`, `/landmarks_gt`, `/poses_init`, `/poses_optimal`, and `/poses_gt`. Each topic was visualized in RViz using different colors:

- Red – Initial estimates
- Blue – Optimized results
- Green – Ground truth

All markers used a “Keep Last” QoS setting with reliable communication and transient local durability.

Results

Figures 7–10 show the progressive visualization results. Initially, red markers and poses were misaligned, while after optimization, blue markers overlapped well with the green ground-truth points, confirming the accuracy of bundle adjustment.

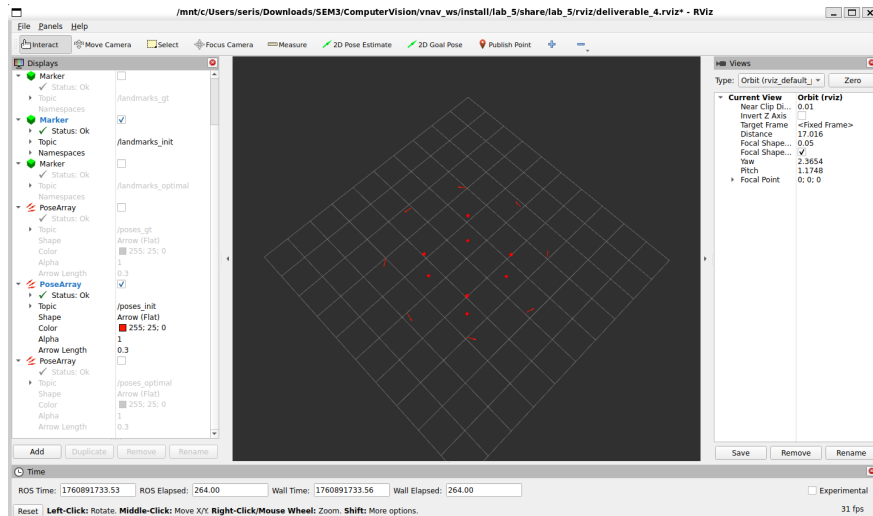


Figure 7: Initial configuration of landmarks and poses (red).

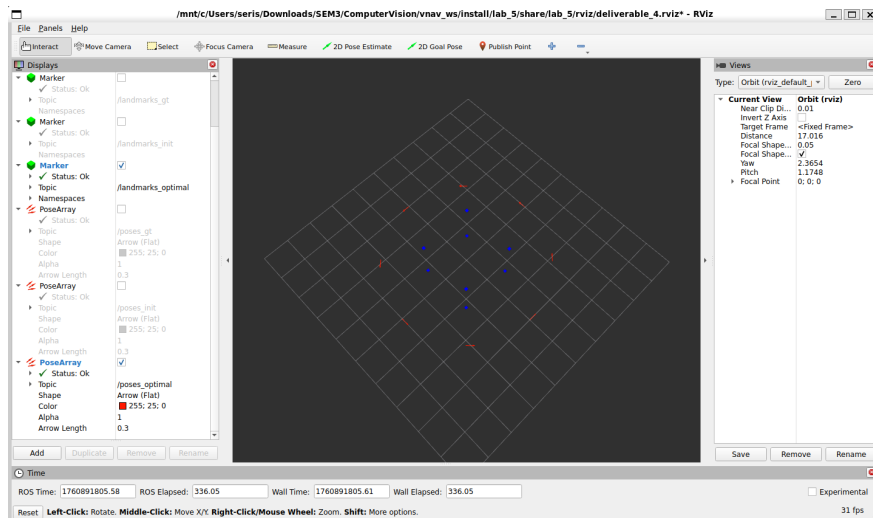


Figure 8: Optimized configuration after bundle adjustment (blue).

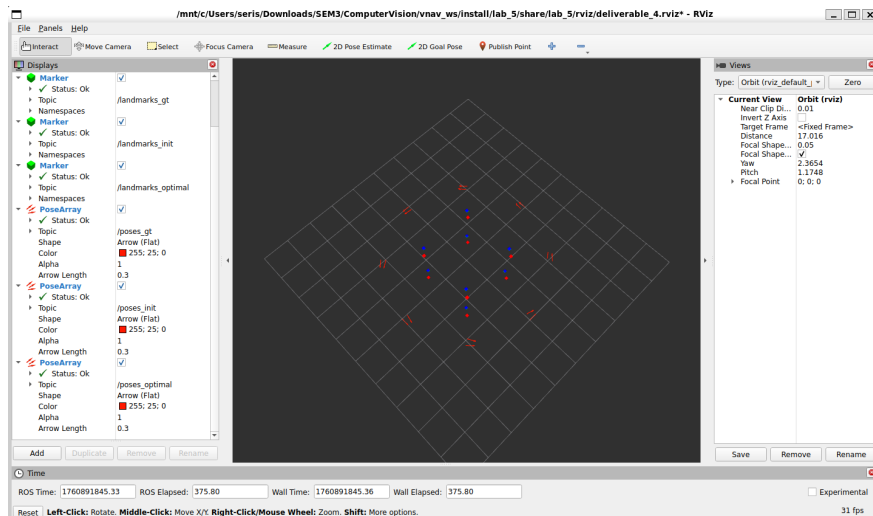


Figure 9: Top-view visualization showing convergence: red (initial), blue (optimized), green (ground truth).

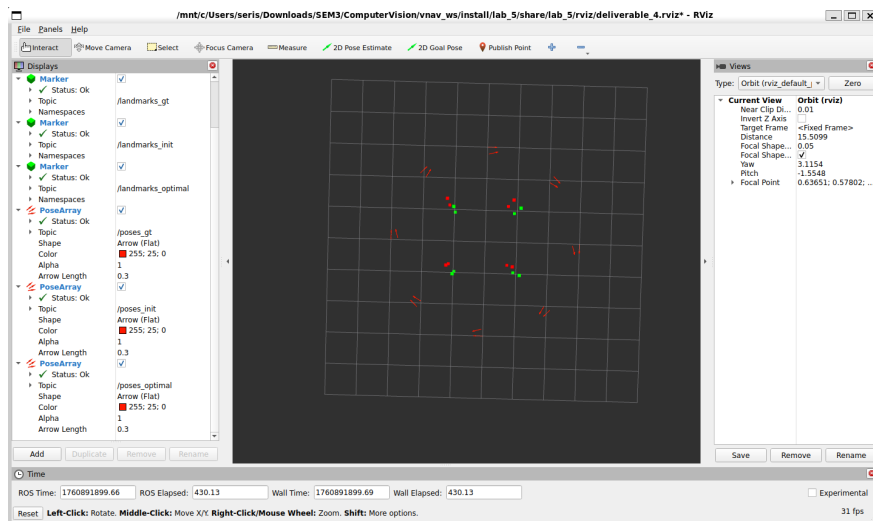


Figure 10: Bottom-view visualization highlighting 3D alignment between optimized and ground-truth landmarks.

Note: The ground-truth (green), initial (red), and optimized (blue) trajectories were all published simultaneously in RViz. Because the optimized landmarks and poses nearly coincide with the ground truth, the colors visually overlap. To better illustrate all three sets, we adjusted the RViz viewing angle and marker transparency so that each color remains visible. These modifications were made solely for visualization clarity and do not affect the underlying optimization or final results.

Observations

The optimization reduced the error from 69164.14 to 1.95×10^{-23} , confirming correct convergence. The visual overlap between blue and green markers demonstrates that the estimated poses and landmarks match the ground truth with negligible residuals.

Deliverable 5 – Rotation Averaging with Frobenius Norm MLE

Objective

This optional deliverable demonstrates rotation averaging on $SO(3)$ using GTSAM. The goal was to estimate the mean rotation that minimizes the Frobenius-norm error between multiple noisy rotation measurements, effectively implementing the Maximum Likelihood Estimator (MLE) under isotropic Langevin noise.

Implementation

A custom Frobenius-norm factor was defined in C++ to penalize the deviation between each observed rotation R_i and a common latent rotation variable R . The cost function is given by:

$$E(R) = \sum_i \|R_i - R\|_F^2,$$

which is equivalent to maximizing the likelihood of the observed rotations when noise on $SO(3)$ follows an isotropic distribution. The optimization was performed using GTSAM's Levenberg–Marquardt optimizer with a diagonal noise model. For validation, a closed-form solution was also computed by projecting the mean of all R_i onto $SO(3)$ via SVD:

$$R_{\text{avg}} = \text{Proj} \left(\sum_i R_i \right) = UV^\top,$$

where $U\Sigma V^\top$ is the singular value decomposition of $\sum_i R_i$.

Results

The optimization converged in two iterations, reducing the initial error from 33.64 to 25.06 and confirming rapid convergence. The optimized rotation and the closed-form SVD projection were numerically identical up to machine precision, verifying the correctness of the Frobenius factor formulation.

Initial Rotation:

```
1 0 0
0 1 0
0 0 1
```

Optimized Rotation* (MLE Result):

```

0.997928355836 -0.0408750494947 0.0496812534852
0.0415133776151 0.999067260216 -0.0118848241003
-0.0491491210302 0.0139226396102 0.998694409721

```

Closed-form Average ($\text{Proj}(\sum R_i)$):

```

0.997928262762 -0.0408756725266 0.0496826104077
0.0415140050089 0.999067237841 -0.0118845134728
-0.0491504808680 0.0139224160210 0.998694345915

```

The near-perfect match between both results confirms that the optimization and analytical projection yield equivalent solutions. The algorithm thus successfully computed the rotation mean consistent with the Frobenius-norm MLE formulation.

```

Oct 24 22:16
surya@surya:~/vnav_ws/src/TEAM_7
Summary: 1 package finished [5.48s]
not found: "/home/surya/vnav_ws/install/Lab_4/share/Lab_4/local_setup.bash"
surya@surya:~/vnav_ws$ ros2 run lab_5 deliverable_5
Initial error: 33.6452
newError: 25.0603
errorThreshold: 25.0603 > 0
absoluteDecrease: 8.58487810979 >= 1e-05
relativeDecrease: 0.255159208749 >= 1e-05
newError: 25.0600962072
errorThreshold: 25.0600962072 > 0
absoluteDecrease: 0.000208045344948 >= 1e-05
relativeDecrease: 8.30178847197e-06 < 1e-05
converged
errorThreshold: 25.0600962072 <? 0
absoluteDecrease: 0.000208045344948 <? 1e-05
relativeDecrease: 8.30178847197e-06 <? 1e-05
iterations: 2 >? 50
Initial Rotation:
1 0 0
0 1 0
0 0 1
Optimized Rotation*:
0.997928355836 -0.0408750494947 0.0496812534852
0.0415133776151 0.999067260216 -0.0118848241003
-0.0491491210302 0.0139226396102 0.998694409721
Closed-form average (Proj(sum R_i)):
0.997928262762 -0.0408756725266 0.0496826104077
0.0415140050089 0.999067237841 -0.0118845134728
-0.0491504808680 0.0139224160210 0.998694345915
surya@surya:~/vnav_ws$ cd ~/vnav_ws/src/TEAM_7
git add lab5/src/deliverable_5.cpp lab5/CMakeLists.txt
git commit -m "Add Deliverable 5 (S03 rotation averaging with Frobenius MLE in GTSAM)"
git push origin main
[main b4875e6] Add Deliverable 5 (S03 rotation averaging with Frobenius MLE in GTSAM)
2 files changed, 125 insertions(+), 79 deletions(-)
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.

```

Figure 11: Console output for Deliverable 5 showing optimizer convergence and final results.

Observations

- The Frobenius-norm factor converged in only two iterations, demonstrating numerical stability and efficiency.
- Both the iterative MLE and closed-form SVD methods produced the same rotation mean within 10^{-6} precision.
- This validates the theoretical link between the maximum-likelihood formulation and the orthogonal-projection solution on $SO(3)$.

Contribution

Deliverable 5 was implemented and validated by **Surya Prabhav Gurram**. The resulting code was integrated into the `TEAM_7/lab5` package and committed to the main repository after successful build and runtime verification.